

Video Vanguard

With the demise of 3dfx, ATi and nVidia are the only high-end graphics chip manufacturers with significant market domination.

Mainstream desktop display solutions starting from Rs 1,500 to Rs 40,000 are now available in the market with a great number of manufacturers offering a wide variety of solutions in the lower segments. With an increase in competition, the choices available have now improved significantly.

Technology Insight

The main aim of a graphics accelerator is to process the 2D or 3D elements associated with the GUI and games or software applications. The execution of graphics involves the transfer of texture maps, image geometry, commands etc., over to the accelerator for processing.

Smart Tips

Graphics processor: The core of a graphics card, this is the component that actually processes the video information. Some of the best graphics processors in the market are the

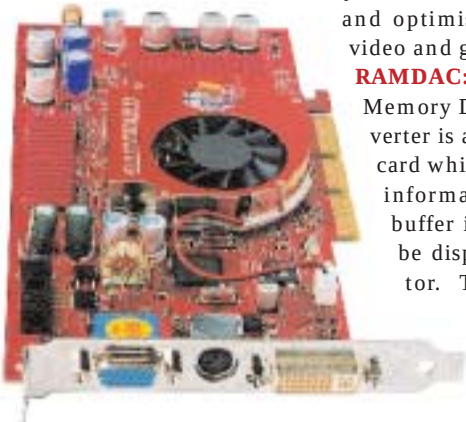
nVidia GeForce FX and the ATi Radeon 9800 Pro.

Video RAM: Video RAM on the graphics card is used to store the extensive graphical information, such as textures of 3D objects. It is also used as the frame buffer—the part that holds the final frame that is to be displayed on the monitor. This RAM is very similar to system RAM, except that it is faster and optimised for working with video and graphical information.

RAMDAC: The Random Access Memory Digital to Analog Converter is a chip on the graphics card which converts the digital information in the frame buffer into a signal that can be displayed on your monitor. The RAMDAC on a graphics accelerator is specified by a frequency. The higher the frequency of the RAMDAC, the

greater the resolutions and refresh rates that can be supported by the card.

Interface: The best choice available today is the AGP interface, which provides a very high-speed pathway for moving large quantities of graphic and image data between the graphics card and the rest of the system.



Also Remember...

■ All video cards use a VGA output connector that allows the signal to be displayed on a monitor. Some cards supply and receive the video signal from television monitors, projectors and video cameras through optional coaxial or S-video interfaces.

■ Some cards have a feature connector which supports devices such as a TV tuner or a video capture board. A special flat ribbon connector is used to connect the video card to these additional devices.

■ The memory bandwidth refers to the rate at which data is transferred between the graphics processor and graphics memory. The effect that memory speed has on the total memory bandwidth and the importance of it, can be understood by looking at the following examples:

GeForce FX5800:
128-bit (16 bytes) x 800 MHz
= 16 x 800 = 12.81 GBps

Geforce2 Ultra:
128-bit (16 bytes) x 460 MHz
= 16 x 460 = 7.36 GBps

TNT2 M64:
64-bit (8 bytes) x 125 MHz
= 8 x 125 = 1 GBps

Decision Maker

	Limited budget	Should run most games, but price is a consideration	Top-notch performance in 2D/3D, price no
You need	A card that will run at 32-bit colour and 1,024 x 768 at 85 MHz with support for the odd game at low resolutions	A card that will give you a minimum of 30 fps in low to medium resolutions with little compromise in quality	Maximum performance with all the eye candy enabled and no compromise on either resolution or quality
Look for	Cheaper or lower variants of faster chips. A card using the AGP interface with minimum 16 MB onboard RAM	A card with at least 32 MB or more memory	A card with 64 MB RAM or more with features such as dual-monitor support
Our pick	Riva TNT2	GeForce2 MX 200	ATi Radeon 9700 Pro, Geforce FX 5800
Price range	Up to Rs 1,500	From Rs 2,500 to Rs 5,000	Rs 36,000 onwards